

TRANSducers

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Introduction

An electronic instrumentation system consists of a number of components which together are used to perform a measurement and record the result. An instrumentation system generally consists of three major elements: an input device, a signal-conditioning or processing device, and an output device. The input device receives the quantity under measurement and delivers a proportional electrical signal to the signal-conditioning device. Here the signal is amplified, filtered, or otherwise modified to a format acceptable to the output device. The output device may be a simple indicating meter, an oscilloscope, or a chart recorder for visual display. It may be a magnetic tape recorder for temporary or permanent storage of the input data, or it may be a digital computer for data manipulation or process control. The final output depends on what is to be measured and how the measurement result is to be presented.

The input quantity for most instrumentation systems is non-electrical. In order to use electrical methods and techniques for measurement, manipulation or control, the non-electrical quantity is converted into an electrical signal by a device called a transducer. Thus transducer is defined as a device which, when actuated by energy in one transmission system, supplies energy in the same form or in another form to a second transmission system. "The energy transmission may be electrical, mechanical, chemical, optical (radiant), or thermal."

② A transducer may be defined as a device that convert mechanical force or displacement into an electrical signal.

Many physical parameters such as heat, light intensity, humidity may also be converted into electrical energy by means of transducers.

CLASSIFICATION OF TRANSDUCERS

Transducers may be classified according to their application, method of energy conversion, nature of the output signal, etc.

Transducers are classified according to the electrical principles involved; passive transducers and active transducers.

ELECTRICAL TRANSDUCERS

Electrical transducers convert non-electrical quantities into electrical quantities first and then measured.

Primarily quantities to be measured are non-electrical such as temperature, pressure, displacement, humidity, fluid flow, speed etc.

TYPES OF ELECTRICAL TRANSDUCERS

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Various types of electrical transducers are as follows:
(i) Resistive type (ii) Capacitive type (iii) Inductive type.

RESISTIVE TYPE TRANSDUCER

In such a transducer, resistance between the output terminals of a transducer gets varied according to the measurand. Resistive transducers are performed over other transducers because dc and ac both are suitable for resistance measurement.

Resistance of any metal conductor is given by the equation

$$R = \rho \frac{L}{a}$$

... (1)

where ρ is the resistivity of the material of conductor in $\Omega \cdot m$, L is the length of conductor in meters, and a is the cross sectional area of conductor in m^2 .

Physical phenomenon of input signal to the transducer causes variation in resistance by changing any one of the quantities ρ , L and a . For measurement of displacement length of conductor is varied in potentiometer thereby resulting in change in resistance.

For measurement of force and pressure resistance of a conductor or semiconductor is varied in strain gauge. Variation in temperature causes change in the resistivity of the conductor material and so change in resistance takes place which is used for measurement of temperature. With some devices resistance varies with the change in light intensity because of photoconductive effect, while with other resistance varies on exposure to magnetic field due to magnetoresistive effect. Thus resistive transducers have a wide field of application. Most resistive transducers lies in the mid-range 100 Ω - 100 k Ω .

INDUCTIVE TRANSDUCEES

(4)

These are analog passive transducers. These transducers operate generally upon one of the following three principles.

- (i) Variation of Self inductance of the coil
- (ii) Variation of Mutual inductance of the coil.
- (iii) Production of eddy currents.

1. Transducers operating on the principle of Variation of Self inductance.

The self inductance of a coil is given by the equation

$$L = \frac{N^2}{l/\mu A} = N^2 \frac{\mu A}{l} = N^2 \mu G \quad \text{--- (2)}$$

Where N is the number of turns on the coil, l is the mean length of the magnetic path, A is the area of the cross-section of the magnetic path and μ is the permeability of the magnetic materials. μ is called the geometric form factor; G .

From the above equation it obvious that the self inductance of a coil can be varied by varying the number of turns on the coil, the permeability of the magnetic material or by changing the geometrical configuration of the magnetic circuit.

These transducers are usually used for measurement of displacement (linear as well as angular). The displacement under measurement causes change in self inductance of the coil L by ΔL by varying any of the three variables N^2 , μ or G .

2- Transducers operating on the principle of variation of ⁽⁵⁾ Mutual inductance.

Such transducers operate on the fact that mutual inductance between two coils depends upon the self inductances of the coils and coefficient of coupling between them, as mutual inductance between two coils is given by the expression

$$M = K \sqrt{L_1 L_2}$$

--- (3)

Where K is the coefficient of coupling and L_1 and L_2 are self inductances of the coils.

In such transducers two or more coils are used and the mutual inductance between the coils is varied as per requirement.

A simple application of inductive transducer using this principle for measurement of linear displacement is shown in figure 6. In such a transducer, both the coils are in series so have inductance of both the coils as an algebraic sum of self inductance of both coils and mutual inductance between the coils.

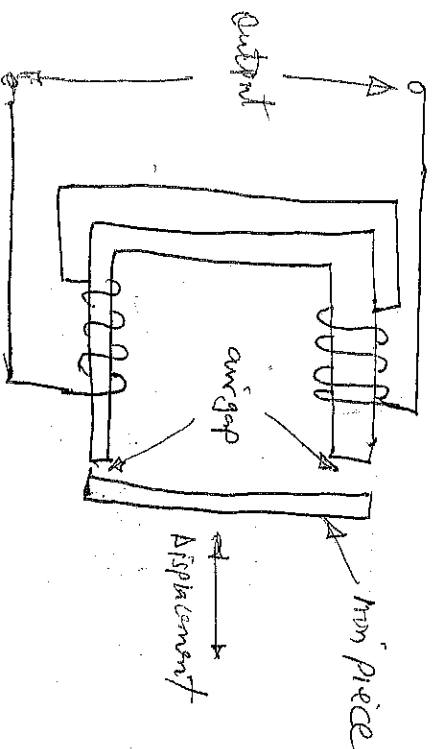


Figure 6: Inductive transducer for measurement of linear displacement

The inductance of such an arrangement varies from $L_1 + L_2 - 2M$ to $L_1 + L_2 + 2M$.

With displacement, moving iron piece causes change in wiring of the magnetic circuit, so change in magnetic flux linking with the coils resulting change in mutual inductance between the coils. This change in mutual inductance is sensed and indicated by the bridge circuit.

An arrangement for measuring angular displacement is shown in Figure 7. It consists of two air-cored coils—one moving and another fixed one.

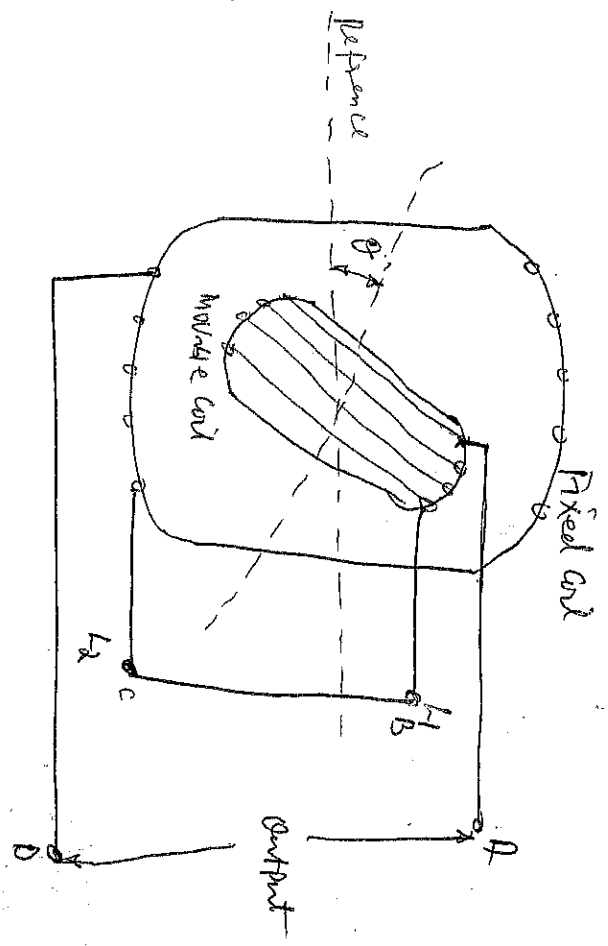


Figure 7: Inductive transducer for measurement of angular displacement

The movable coil is made to rotate by the input angular displacement θ . The self inductance of each coil is constant but the mutual inductance between the two coils varies on account of variation of angle θ . The coefficient of coupling is proportional to $\cos \theta$. The mutual inductance is maximum when the axis of two coils coincide. The inductance between terminals A and B varies from $L_1 + L_2 - 2M$ to $L_1 + L_2 + 2M$.

3. Transducers operating on the principle of production of eddy currents ⑦ *

These transducers operate on the principle that when a conducting plate is kept near a coil carrying alternating current, eddy currents are induced in the conducting plate, producing its own magnetic field in opposition to the main field created by the coil. Thus ~~eddy currents~~ induced in the conducting plate reduce the net flux linking with the coil and so the inductance of the coil is reduced. The closer is the plate to the coil, the higher are the induced eddy currents and so higher is the reduction in the inductance of the coil. Thus, the inductance of the coil changes with the movement of the plate.

An inductive transducer based on this principle is shown in Figure 8.10.

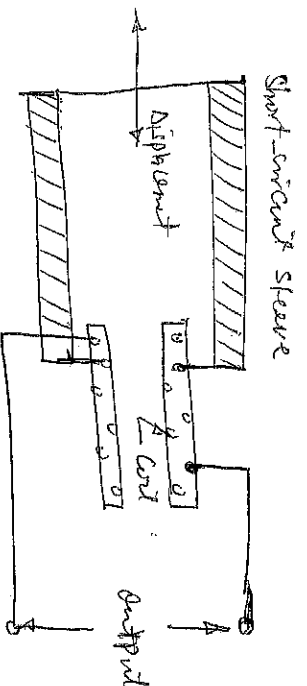


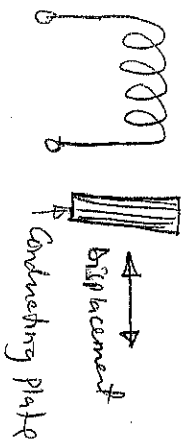
Figure 8.10 Inductive transducer for measurement of linear displacement

In such a transducer, there is a coil carrying an alternating current and a cylindrical sleeve of some ferrous conducting material moving parallel and coaxially to the coil. When the sleeve covers the length of the coil, low eddy currents are induced in it so more is the inductance of the coil and when the sleeve covers more length of the coil, more eddy currents are induced in it and so less is the inductance of the coil. This change in inductance, the measure of displacement, is sensed and indicated by an

as indicated by an ac bridge circuit.

(8) *

In another arrangement, the plate may be placed at right angle to the axis of the coil, as shown in Figure 9(b). The displacement of the plate causes a change in the coil inductance.



(b)

Sensitivity of this type of inductive transducer is usually low.

CAPACITIVE TRANSDUCERS

The variable capacitance transducer comprises of a capacitor, the capacitance of which is varied by the non-electrical quantity being measured.

The capacitance of a parallel plate capacitor, C is given by an

equation
$$C = \frac{\epsilon_0 \epsilon_r A}{d} \text{ Farads} \quad \dots (4)$$

where ϵ_0 is the permittivity of free space and is equal to $8.85 \times 10^{-12} \text{ FM}$; ϵ_r is the relative permittivity of the dielectric, A is the area of plates in m^2 and d is the distance between the two plates in mtrs.

As the capacitance of a capacitor depends upon the area of its electrodes, the distance between them and the relative permittivity of the dielectric material, hence such transducers can be used for measurement of non-electrical quantities by affecting any one of the above mentioned parameters.

Capacitive transducers are analog passive transducers.

In such transducers capacitance of the capacitor is varied by any of the following three methods;

⁽¹⁰⁾ Capacitive Transducer - By variation of overlapping area of plates.

Such a transducer operates on the fact that any capacitor is proportional to the overlapping area of plates.

Consider a capacitive transducer having parallel plates of constant width b , as shown in figure below. In such a transducer, capacitance is proportional to l where l is the length of overlapping portion of the plates and it varies according to the displacement and measurement. So any change in displacement causes change in capacitance. For a parallel plate capacitor, the capacitance C is given by an expression:

$$C = \frac{\epsilon_0 \epsilon_r A}{d} = \frac{\epsilon_0 \epsilon_r b l}{d} \text{ Farads}$$

where l is the length of overlapping portion, b is the width of plates in meter and d is separation between plates in meter.

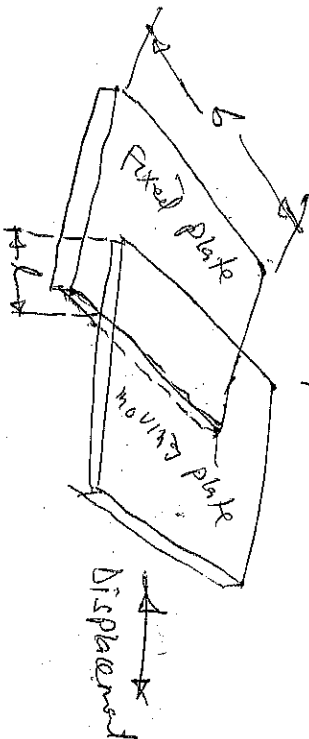


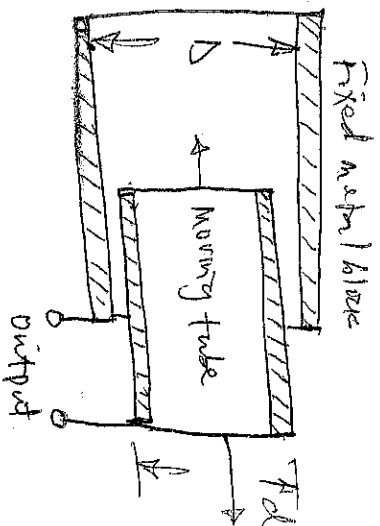
Figure Capacitive transducer by variation of overlapping area of plates

where l is the length

Sensitivity, $S = \frac{\delta C}{\delta l} = \frac{\epsilon_0 \epsilon_r b}{d} \text{ F/m}$

(11)

For a cylindrical capacitor where overlapping is varied by varying the length of overlapping of the cylinders, as shown in Figure



Figure

For a cylindrical capacitor shown in figure above, the capacitance is given by the expression

$$C = \frac{2\pi\epsilon_0\epsilon_r l}{\log_e \frac{D}{d}} \quad \text{Farads} \quad \dots (C)$$

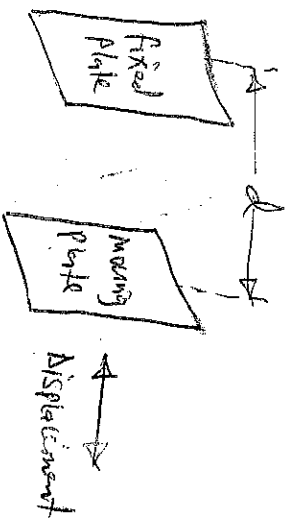
Where D is the inner diameter of outer cylindrical electrode in meters, d is the outer diameter of inner cylindrical electrode in meters and l is the length of overlapping portion of cylinders in meters.

$$\text{Sensitivity, } S = \frac{\delta C}{\delta l} = \frac{2\pi\epsilon_0\epsilon_r}{\log_e D/d} \quad \text{p/m} \quad \dots$$

So So

CAPACITIVE TRANSDUCERS - by variation of distance between plates. (18)

This type of transducer operates upon the fact that the capacitance of any capacitor is inversely proportional to the distance between the plates. Such a capacitive transducer used for measurement of linear displacement as shown in Figure below



Figure

This transducer consists of one plate fixed and other moving. As the moving plate moves away from or towards the fixed plate as per displacement under measurement, the capacitance decreases or increases. This change in capacitance is measured by an ac bridge circuit or with an oscillating circuit and this displacement of the moving plate is determined.

From the expression for capacitance of a parallel plate capacitor, small change in capacitance due to small displacement is given as

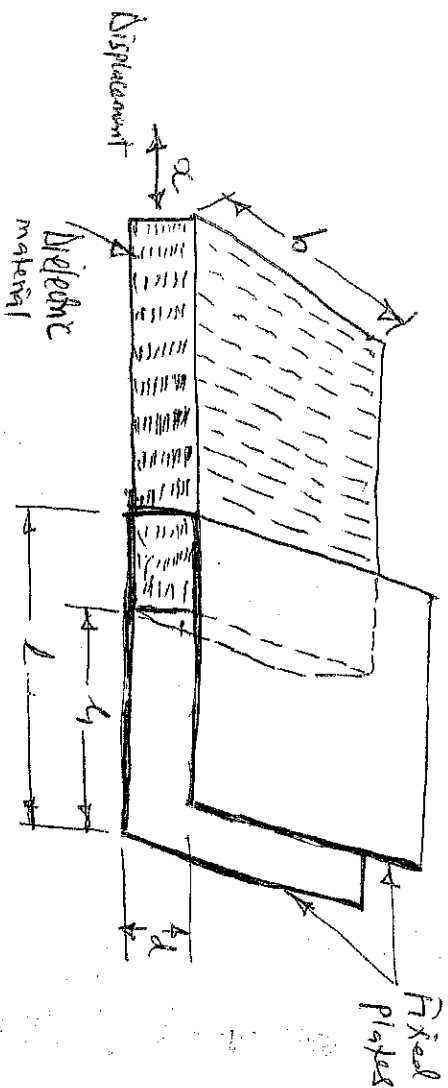
$$\frac{\delta C}{C} = \frac{-\epsilon_0 \epsilon_r A}{d^2}$$

From the above expression it is obvious that the sensitivity, being equal to $\frac{\delta C}{C}$ or $\frac{-\epsilon_r A}{d^2}$ is not constant by varies

over the transducer range.

CAPACITIVE TRANSDUCER — by variation of permittivity of the dielectric material between the plates. (18)

In such a transducer, capacitance of the transducer is varied by varying the permittivity of the dielectric material between the plates.



Figure

In this arrangement, a dielectric material of relative permittivity ϵ_r moves between the two fixed parallel plates according to the displacement under measurement. The capacitance of the transducer is given by

$$C = \epsilon_0 \frac{b^2 l}{d} + \epsilon_0 \epsilon_r \frac{b(L-l)}{d}$$

If dielectric material moves and a distance x , then the value of capacitance changes from C to $C + \Delta C$, and is given by the expression

$$\Delta C = \frac{\epsilon_0 b^2 x}{d} (\epsilon_r - 1)$$

So change in capacitance is proportional to the displacement. This method can be used to measure displacement from $1 \mu m$ to $10 mm$. (19)

Example

(14)

A capacitance transducer of two parallel plates of overlapping area of $5 \times 10^{-4} \text{ m}^2$ is immersed in water. The capacitance C has been found to be 9.5 pF . Calculate the separation d between the plates and the sensitivity, $S = \frac{dC}{d}$ of the transducer. Given ϵ_r for water = 81; $\epsilon_0 = 8.854 \text{ pF/m}$.

Soln: -

$$\text{Since } C = \frac{\epsilon_0 \epsilon_r A}{d}$$

$$\text{Separation between plates, } d = \frac{\epsilon_0 \epsilon_r A}{C} = \frac{8.854 \times 10^{-12} \times 81 \times 5 \times 10^{-4}}{9.5 \times 10^{-12}}$$

$$= 37.75 \times 10^{-3} \text{ m or } 37.75 \text{ mm}$$

$$\text{Sensitivity, } S = \frac{dC}{d} = \frac{-\epsilon_0 \epsilon_r A}{d^2} = \frac{-8.854 \times 10^{-12} \times 81 \times 5 \times 10^{-4}}{(37.75 \times 10^{-3})^2}$$

$$= 0.025 \times 10^{-8} \text{ F/m}$$

Example

A capacitive transducer is made up of two concentric cylindrical electrodes. The length of electrodes is 25 mm , the inner diameter of outer cylindrical electrode is 4.8 mm and the outer diameter of inner cylindrical electrode is 4.0 mm . Determine the sensitivity of the transducer. Assume air medium.

Determine the change in capacitance for a displacement

(20)

of the inner electrode of 2.5 mm.

(15)

Determine also the electric stress when a voltage of 150V is applied across the electrodes.

Soln: -

Electrode length, $l = 2.5 \text{ mm} = 0.025 \text{ m}$

Inner diameter of outer electrode, $D = 4.2 \text{ mm} = 4.2 \times 10^{-3} \text{ m}$

Outer diameter of inner electrode, $d = 4.0 \text{ mm} = 4 \times 10^{-3} \text{ m}$

Relative permittivity of medium (air) $\epsilon_r = 1.0$

Transducer Sensitivity, $S = \frac{\delta C}{\delta l} = \frac{2\pi \epsilon_0 \epsilon_r}{\log_e D/d} \text{ F/m}$

$$= \frac{2\pi \times 8.854 \times 10^{-12} \times 1}{\log_e \frac{4.2}{4.0}} = 1140 \text{ pF/m}$$

or 1.14 pF/mm

Change in capacitance for a displacement of the inner electrode of 2.5 mm

$$\Delta C = S \times \Delta l = 1.14 \times 2.5 = 2.8 \text{ pF}$$

$$\begin{aligned} \text{length of air gap between two electrodes} &= \frac{D-d}{2} = \frac{4.2 - 4.0}{2} \\ &= 0.1 \text{ mm} \end{aligned}$$

Applied voltage across the electrodes, $V = 150 \text{ V}$

$$\begin{aligned} \text{Dielectric stress} &= \frac{V}{\text{Air gap length}} = \frac{150}{0.1} = 1500 \text{ V/mm} \end{aligned}$$

(21)